

APPLICATION OF ADSORPTION THEORIES TO DESCRIBE CEFOPERAZONE PRECONCENTRATION USING AMINATED SILICA

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The use of theoretical sorption models makes it possible to describe the equality in the system and calculate the physico-chemical characteristics to optimize the conditions and understand the mechanisms of interaction. In this work, the application of sorption theories to the preconcentration of cefoperazone (Cefoper) with silica modified with polydimethyldialammonium (PDDA), polyhexamethylene guanidine (PGMG) and polyethylenimine (PEI) was studied.

Cefoper is quantitatively (98%) extracted from aqueous solutions with sorbents SiO₂-PDDA, SiO₂-PGMG at pH 3-6. The experimentally determined sorption capacity of SiO₂-PDDA, SiO₂-PGMG, and SiO₂-PEI according to Cefoper is 0.118, 0.087, and 0.104 mmol/g, respectively. The experimental sorption isotherms are satisfactorily described by the Langmuir model (see the table). The adsorption equilibrium constant K_L decreases with the transition from SiO₂-PDDA and SiO₂-PGMG to SiO₂-PEI, which is associated with the greater strength of the interaction of Cefoper with PGMG and PDDA, and this is confirmed by experimental data. The values of the coefficients R_L of the Langmuir model and n of the Freundlich model indicate active sorption. The energy of interaction between Cefoper and sorbent polyamines decreases as the surface is filled. The free energy according to the Dubinin-Radostkevich model $E < 8$ kJ/mol indicates the physical mechanism of sorption, the energy according to the Temkin (B) model indicates the exothermic nature of the process.

Parameters of theoretical models of Cefoperazone sorption

The sorption model	Parameter	SiO ₂ - PDDA	SiO ₂ - PGMG	SiO ₂ - PEI
Langmuir	$a_{\max}$, mmol/g	0.134	0.104	0.192
	K_L , l/mmol	11.0	9.28	1.91
	R_L	0.1-0.9	0.1-0.9	0.3-1.0
	R^2	0.993	0.994	0.757
Freundlich	K_F , mmol/g	0.258	0.162	0.211
	$1/n$	0.662	0.617	0.888
	R^2	0.933	0.963	0.943
Dubinin-Radostkevich	$a_{\max}$, mmol/g	0.129	0.098	0.121
	K , mol ² /kJ ²	0.017	0.017	0.032
	E , kJ/mol	5.36	5.44	3.92
	R^2	0.984	0.987	0.995
Temkina	A_T , l/mg	209	165	54
	B , kJ/mol	23.5	18.4	27.7
	R^2	0.922	0.937	0.934